Simple Bank Application Project

1. Objective

Build a simple banking system where users can:

- Create an account

- View account details

- Deposit money

- Withdraw money

- View transaction history

This project helps students understand: - MVC architecture - REST APIs - Database design (SQL) - Basic frontend-backend integration

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2. Tech Stack (Suggested)

- Backend: Node/ Spring Boot (Java)/ Python
- Frontend: React with basic HTML, CSS, JavaScript
- Database: MySQL
- Tools: VS Code, Postman, Swagger

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3. High-Level Architecture

Frontend (UI) → REST API (Controller) → Service Layer → Repository → Database

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4. Database Design

4.1 Tables

USERS

```sql
CREATE TABLE users (
    user_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    email VARCHAR(100) UNIQUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

ACCOUNTS

```sql
CREATE TABLE accounts (
    account_id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT,
    balance DECIMAL(10,2) DEFAULT 0,
    account_type VARCHAR(50),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(user_id)
);
```

TRANSACTIONS

```sql
CREATE TABLE transactions (
    txn_id INT PRIMARY KEY AUTO_INCREMENT,
    account_id INT,
    txn_type VARCHAR(20),
    amount DECIMAL(10,2),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (account_id) REFERENCES accounts(account_id)
);
```

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5. Backend Design (MVC)

5.1 Model (Entities)

- User
- Account
- Transaction

5.2 Repository Layer

- UserRepository
- AccountRepository
- TransactionRepository

5.3 Service Layer (Business Logic)

AccountService Methods:

- createAccount(userId, accountType)
- getAccount(accountId)
- deposit(accountId, amount)
- withdraw(accountId, amount)
- getTransactions(accountId)

5.4 Controller (REST APIs)

Create Account

POST /api/accounts

```
{
  "userId": 1,
  "accountType": "SAVINGS"
}
```

Get Account Details

GET /api/accounts/{id}

Deposit Money

POST /api/accounts/{id}/deposit

```
{
  "amount": 500
}
```

Withdraw Money

POST /api/accounts/{id}/withdraw

```
{
  "amount": 200
}
```

Transaction History

GET /api/accounts/{id}/transactions

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6. Business Rules

- Cannot withdraw more than balance
- Deposit amount must be positive
- Maintain transaction record for each deposit/withdraw

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7. Frontend (Basic UI Screens)

7.1 Home Page

- Buttons:
  - Create Account
  - View Account

7.2 Create Account Page

Fields: - Name - Email - Account Type (Dropdown)

Button: Submit

7.3 Account Details Page

Display: - Account ID - User Name - Balance

Buttons: - Deposit - Withdraw - View Transactions

7.4 Deposit Page

- Input: Amount
- Button: Submit

7.5 Withdraw Page

- Input: Amount
- Button: Submit

7.6 Transaction History Page

Table: - Transaction ID - Type (Deposit/Withdraw) - Amount - Date

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8. Sample JSON Responses

Account Response

```
{
  "accountId": 1,
  "userName": "John Doe",
  "balance": 1000.00
}
```

Transaction Response

```
[
  {
    "type": "DEPOSIT",
    "amount": 500,
    "date": "2026-03-20"
  }
]
```

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9. Bonus Enhancements (Optional)

- Add login/authentication
- Add transfer between accounts
- Add pagination for transactions
- Add validation messages on UI

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10. Evaluation Criteria

- Correct API implementation
- Clean MVC separation
- Working UI flows
- Proper database usage

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11. Submission Requirements

- Source code (GitHub)
- SQL script
- Screenshots of UI
- Postman collection

This project is intentionally simple and designed for beginners to understand full-stack development basics.